

# Habibul Rahman Qalbi

## AI Engineer · Backend Engineer · Full Stack Developer

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Open to remote roles worldwide, hybrid / on-site in Indonesia, or relocation (NZ, AU, SG, MY, TW, JP, US).

### SUMMARY

Software engineer with 4+ years of backend-heavy experience, from full stack web and IoT systems to production AI. Designs and ships agentic AI and Retrieval-Augmented Generation (RAG) services in Python (FastAPI, Django), and React / Next.js frontends. Track record of cutting AI service costs by up to 70% while improving performance up to 4×, and working directly with C-level leaders to turn ambiguous requirements into reliable products.

### SKILLS

|           |                                                                                                                                                     |
|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| Languages | Python, TypeScript, JavaScript, C++                                                                                                                 |
| AI & LLM  | OpenAI, Anthropic, Google GenAI, VoyageAI, LangChain, LlamaIndex, Pydantic AI, agentic workflows, RAG, tool calling, structured outputs, TensorFlow |
| Backend   | FastAPI, Django, Django REST Framework, Celery, Django Channels, REST APIs, WebSockets, JWT, MQTT                                                   |
| Frontend  | React, Next.js, Redux Toolkit, RTK Query, Tailwind CSS, Material UI, Three.js                                                                       |
| Data      | PostgreSQL, MongoDB, Redis, Pinecone, ChromaDB, Apache Airflow                                                                                      |
| DevOps    | Docker, CI/CD, Git, Azure, RabbitMQ, MinIO, Playwright                                                                                              |

### EXPERIENCE

AI / LLM Engineer

May 2024 – Present

Airconcur (Growise) · Taipei, Taiwan · Remote

- Architect and ship modular agentic and RAG-based AI microservices in Python (FastAPI) using multiple LLMs and embedding models, tailored per client.
- Work on TAM AI, an AI go-to-market platform: agent architecture (multi-step reasoning, tool calling, structured outputs), task management and infrastructure decisions.
- Delivered AI services for enterprise customers: document generation, conversational assistants, AI transcription and semantic image search.
- Reduced AI service operating costs by up to 70% and improved performance up to 4× through prompt optimization, retrieval tuning and logic refinement.
- Integrated web scraping and Playwright-based browser automation into agent workflows; improved CI/CD for 2× faster builds.

AI Module Developer (Freelance)

Nov 2023 – May 2024

Airconcur (Growise) · Remote

- Built containerized RAG modules with multi-LLM and multi-embedding support, reusable across application domains.
- Designed isolated, per-user index pipelines to prevent cross-user data leakage; cut retrieval latency by about 30% and raised reasoning accuracy by 60%.

Full Stack Developer / Software Engineer

May 2023 – May 2024

PT Mandiri International Technology · South Jakarta, Indonesia

- Owned the full project lifecycle from requirement gathering to deployment; built customer-facing web apps (React, Next.js) and a cross-platform mobile app on a Django REST backend.
- Engineered IoT device–backend communication over HTTP and MQTT; improved pipeline performance by 50% and cut device bandwidth by 70%.
- Built a site-monitoring system that improved security, logging and performance by 30%; tuned ORM and queries to handle 100+ concurrent requests.

## Full Stack & Backend Developer (Healthcare AI, Freelance)

Aug 2022 – Mar 2023

*Institut Teknologi Sepuluh Nopember (ITS) · Surabaya, Indonesia*

- Built a React–Django app with a U-Net (TensorFlow) backend for deep-vein segmentation from ultrasound images and Three.js 3D visualization, supporting DVT diagnosis.
- Added a model handler and preloader to reduce inference latency; replaced legacy auth with JWT and refactored API viewsets.

## Full Stack Web Developer

Apr 2022 – Apr 2023

*LnData Inc. · Surabaya, Indonesia*

- Built React / Django products, secure REST APIs, Celery workers and Apache Airflow pipelines for social-media data platforms with multinational teams.
- Doubled the capabilities of a data-visualization tool; managed deployment servers and resolved security issues.

## EDUCATION & PUBLICATION

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### B.Eng. Computer Engineering

2017 – 2021

*Institut Teknologi Sepuluh Nopember (ITS) · Surabaya, Indonesia · GPA 3.25 / 4.00*

- Qalbi, H. R., Yuniarno, E. M., Rachmadi, R. F. *Klasifikasi Gerakan Cuci Tangan Berbasis Convolutional Neural Network (CNN)*. Jurnal Teknik ITS, 10(2), 2021. doi:10.12962/j23373539.v10i2.75377

## LANGUAGES

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Indonesian · English

Case studies and demos: [upchunk.github.io](https://upchunk.github.io)